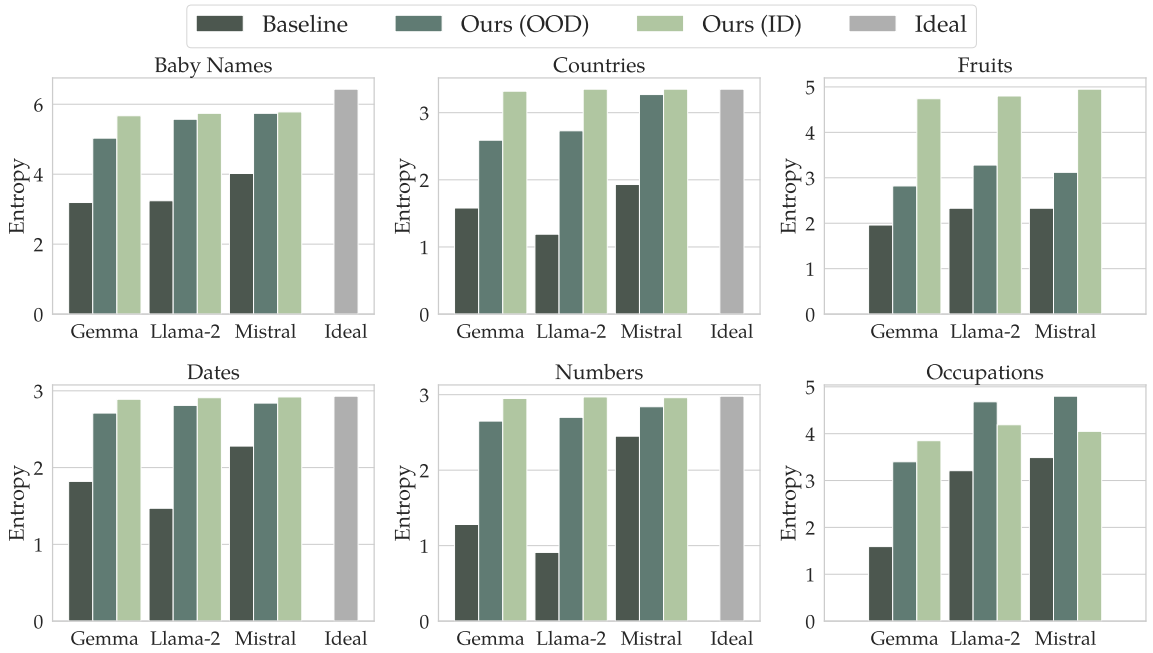


Figure 3: **Entropy in leave-one-out generalization.** The title of each plot indicates which set of tasks we compute entropy over.

We use leave-one-out experiments on a set of six tasks to show that models tuned for diffuse probabilities do, in fact, have strong transferability to tasks unseen during tuning. We consider the following six tasks. Each task has several associated prompts, one of which is shown below:

1. BABYNAMES: "Please generate an English first name, chosen completely at random."
2. COUNTRIES: "Output a random country in Africa, chosen completely at random."
3. FRUITS: "Output a name of a fruit, chosen completely at random."
4. DAYS AND DATES: "Provide a random date in June."
5. NUMBERS: "Randomly pick a prime number between 1 and 50."
6. OCCUPATIONS: "Output an occupation that starts with the letter "A"."

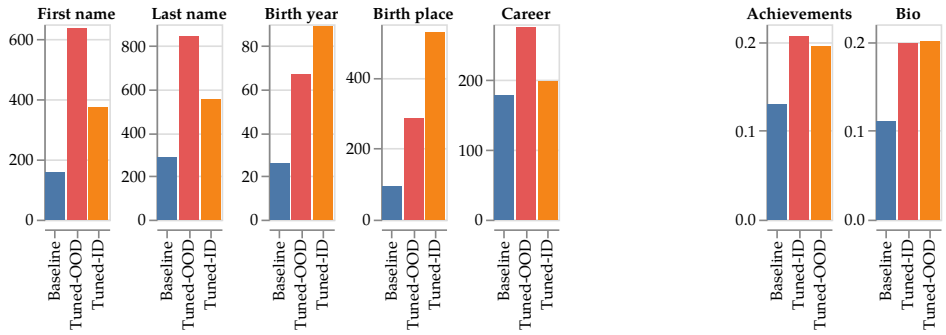
By fine-tuning on five out of the six tasks listed above and evaluating performance on the sixth, we can measure the ability of our method to handle out-of-distribution tasks.

In Figure 3, we report in-distribution (ID) and out-of-distribution (OOD) results, which correspond to tuning sets that include or exclude the particular task, respectively. The results show a convincing trend: for all three models, our tuning method led to substantial improvements in entropy over the baselines, even when the task was held out from the tuning set.[8] Another interesting observation is that the baseline Mistral model consistently produces more diffuse distributions than the baseline Gemma and Llama-2 models, but after tuning, all three models have comparable entropy.

In two of the tasks (COUNTRIES and FRUITS), we observe sizable generalization gaps between in-distribution and out-of-distribution entropy, which suggest that task-specific tuning remains useful, especially when we can come up with a reasonbly diverse set of generation targets for fine-tuning. However, coming up with a large enough target set isn't always easy. In these cases, we rely on the generalization of the models trained on a diverse set of tasks. For example, in OCCUPATIONS, the authors could only come up with a small set of 17 professions that start with the letter "A," and Llama-2 and Mistral (not trained on OCCUPATIONS) generalize out-of-distribution

[8]Coverage results show similar trends, and we report them in Table 5, Appendix C.1.

Figure 4: **Fine-tuned Llama-2 model improves the diversity of synthetic biographies.** We report coverage for categorical attributes in 4a and normalized unigram diversity of generated achievements and the entire biography in 4b.



(a) Coverage results for categorical attributes.

(b) Unigram diversity of biographies.

to occupations beyond what we provide in the fine-tuning set.[9] Notably, our fine-tuning method does not substantially change the general capabilities (e.g., writing and reasoning) of the models demonstrated by evaluations on MT-Bench (see Appendix C.4), making our method compatible with tuning general-purpose large language models.

6 Constructing More Diverse Synthetic Datasets

The leave-one-out experiments show that fine-tuning for diffuse outputs leads to task generalization. This is an important trait for real-world applications such as synthetic dataset construction, where it might be not be feasible to tune on data that is identically formatted to what we would like to synthesize.

In this section, we evaluate how well a model tuned on the six tasks from Section 5 performs on a realistic dataset creation task—building a synthetic dataset of fictional biographies. Inspired by the synthetic datasets created in Maini et al. (2024) and Yuan et al. (2021), these biographies include the following attributes: first and last name, gender, birth year, birth place, profession and a description of the person's achievements. We show how our method results in biographical details that are much more diverse than those generated by the baseline models.

6.1 Out-of-distribution training improves synthetic dataset diversity

We first consider a Llama-2-13B model fine-tuned on all tasks in Section 5 and not specifically for biographies (Tuned-OOD), and compare its generations against the baseline Llama-2 model over 1000 samples. In Figure 4, we report coverage results for categorical attributes (e.g., Birth year), and normalized unigram diversity for achievements and over the entire biography.[10] In Table 2, we report the most frequently generated values for categorical attributes, along with the frequencies out of 1000 generations.[11]

The results on the baseline Llama-2 model indicate that the biases we observe in Section 4.1 towards certain names and numbers expectedly show up in generation. For example, out of 1000 biographies generated by the baseline Llama-2 model, 284 have the first name 'Evelyn,' and 966

[9] For example, Llama-2 generated "Aromatherapist", and Mistral generated "Agronomist." Neither is among the fine-tuning targets of OCCUPATIONS.

[10] We report prompt used for biography generation and side-by-side qualitative examples in Appendix B

[11] A table of coverage and entropy statistics of generated biographies can be found in Appendix C.2.

Table 2: The most frequently generated values for each attribute, along with the number of times the value was generated (out of 1000 generations). Despite being tuned on out-of-domain tasks, the tuned LLaMA significantly improves diversity. *Some values in table have been truncated for brevity.*

First name		Last name		Gender		Birth year		Birth place		Career	
Baseline Llama-2											
Evelyn	284	Nightingale	117	F	966	1985	764	Paris, FR	305	Astronaut	211
Luna	155	Aurora	104	NB	17	1987	46	Tokyo, JP	267	Astro. Engineer	66
Elara	87	Nova	98	M	13	1992	44	Stockholm, SE	33	Aero. Engineer	55
Adriana	42	Starling	53			1978	36	Mumbai, IN	32	Env. Activist	42
Aurora	38	Stardust	41			1975	31	Singapore, SG	32	Astrophysicist	32
Fine-tuned Llama-2 (OOD)											
Luna	32	Nightingale	16	F	762	1985	211	Mumbai, IN	35	Astronaut	96
Zelda	14	Nightshade	12	M	189	1992	99	Lagos, NG	31	Aero. Engineer	50
Mila	14	Chen	8	NB	31	1987	77	Paris, FR	29	Soft. Engineer	47
Evelyn	11	Orion	6			1988	61	Tokyo, JP	27	Env. Activist	35
Althea	9	Sparks	6			1990	52	Nairobi, KE	21	Journalist	34
Fine-tuned Llama-2 (ID)											
Hava	14	Kim	17	F	487	1921	39	Choloma, HN	16	Architect	140
Maria	14	Mohammed	16	M	478	1931	36	Rabat, MA	13	Journalist	74
Juan	14	Khan	13	NB	34	1942	35	Tainan, TW	10	Politician	35
Valcin	13	Abed	13			1916	30	Rajshahi, BD	10	Archaeologist	35
Issaka	12	Salah	12			1984	29	Budapest, HU	10	Mar. Biologist	26

are females. Such a high level of repetition makes the resulting dataset basically unusable without substantial human intervention. In constrast, our model significantly improves the diversity of generated biographies, despite not trained specifically for generating biographies. For example, we see an >2X increase in coverage for most categorical attributes. There are also substantial improvements in generation diversity for achievements and over the entire biography (Figure 4b), although our training does not optimize for open-ended text generation.

The top five most frequent values for attributes (Table 2) help contextualize these improvements in diversity: there are significant reductions in biases towards certain attribute values. E.g., the frequency of the name "Evelyn" decreased by 25X, and the birth year 1985 by over 3X.

6.2 Diversifying distributions of categorical attributes

Despite this improvement over base model, certain biases, as seen in the high frequencies of female biographies (76.2%) and the birth year 1985 (21.1%), still persist, which could limit the utility of the dataset. Our fine-tuning method can in fact be directly applied to balance the distribution of categorical attributes. As a proof of concept, we create a target set containing 210 programmatically generated tuples of categorical attributes, with roughly balanced gender, birth year and birth place distributions without the open-ended achievement descriptions. We then fine-tune a Llama-2 model (Tuned-ID) only on categorical attributes and evaluate another sample of 1000 biographies.

At a comparable level of generation diversity to the Tuned-OOD model (Figure 4), the Tuned-ID model is able to generate biographies with much more balanced distributions of categorical attributes: Table 2 shows that both gender and birth years are roughly uniformly distributed, and a wider range of birth places are produced by the model.[12] Crucially, the model remains highly diverse on open-ended generation of achievements even when being trained exclusively on categorical attributes. This result highlights the potential in extending our fine-tuning method towards improving diversity in open-ended text generation.

[12]See maps of generated birth places for all three models in Figure 5, Appendix C.